Ethiopian Association of Anesthetists

Clinical Audit

Facilitator Guide

December, 2024

Addis Ababa, Ethiopia

Document Management Control

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Date issued: June 2023

Last review: December 2024

Next review: June 2025

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Acknowledgement

The development, review, and approval of this Clinical Audit Continuous Professional Development Training Module would not have been possible without the collaborative efforts and dedication of multiple stakeholders. We extend our deepest gratitude to the Ethiopian Association of Anesthetists clinical standing committee, our funders for supporting this CPD material development, review and accreditation. Our appreciation goes to the subject matter experts, instructional designers, and reviewers whose expertise and insights enriched the quality and relevance of the module.

We are especially grateful to the Addis Ababa University Continuous Professional Development Directorate for their guidance and approval, ensuring the module meets the highest academic and professional standards. Additionally, we acknowledge the invaluable contributions of healthcare professionals and educators who provided practical feedback during the review process.

Finally, we thank all those who supported this initiative, directly or indirectly, in advancing healthcare quality through effective clinical audit training. Your commitment to excellence has been instrumental in bringing this important resource to execution.

Forward

A clinical audit is a process used to evaluate and improve the quality of care provided to patients. It involves analyzing patient data and identifying areas where improvements can be made, and then implementing changes to address those areas. The goal of a clinical audit is to ensure that patients receive safe, effective, and efficient care. The results of a clinical audit can be used to inform practice and policy changes, as well as to identify areas for further research. The process of clinical audit can be forward by having a clear aim, selecting appropriate sample, data collection, data analysis, feedback, and implementation of changes.

Approval statement

As the Director of Continuous Professional Development (CPD) at Addis Ababa University, it is my privilege to endorse the Clinical Audit Continuous Professional Development Training Module. This manual represents a comprehensive and meticulously developed resource aimed at equipping healthcare professionals with the knowledge, skills, and competencies essential for conducting effective clinical audits. The training module aligns with our country's commitment to advancing evidence-based practices and fostering a culture of continuous improvement in healthcare delivery.

The manual has been carefully reviewed and approved for its academic rigor, practical relevance, and alignment with national and institutional priorities. Its structured approach, incorporating core competencies, systematic methodologies, and practical tools, ensures that participants can confidently plan, implement, and sustain impactful clinical audits in their respective settings. By addressing existing gaps and promoting measurable improvements in healthcare quality, this CPD module stands as a vital resource for enhancing patient outcomes and strengthening health systems.

I commend the Ethiopian Association of Anesthetists and all contributors for their dedication to developing this training manual, and I am confident that it will serve as an invaluable guide for advancing clinical audit practices across Ethiopia.

Dr. Merahi Kefyalew

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Abbreviations and acronyms

CC	Core Competency
CCD	Chief Clinical Director
CPD	Continuous Professional Development
CEU	Continuing Education Unit
TOT	Training of Trainers

1. Introduction

1.1 Background

This clinical audit training is a five days duration training course designed to equip health professionals and healthcare administrators involved in clinical services and working in different areas of health facilities aiming to improve efficiency of clinical service and increased patients' satisfaction on health care delivery.

"Clinical audit can be described as a quality improvement cycle that involves measurement of the effectiveness of healthcare against agreed and proven standards for high quality, and taking action to bring practice in line with these standards so as to improve the quality of care and health outcomes."

Clinical Audit forms the system for improving standards of clinical practice. Aspects of patient care are evaluated against expected standards of care and where necessary, changes are made at an individual, team or service level. A re-audit can then be used to confirm that improvements have been effective. Topics for clinical audit should reflect national and/or local priorities of areas of concerns.

Clinical audit is one of the methods to improve clinical effectiveness and improves patient safety. It is the process of systematic assessment of clinical service against the preset standard. Clinical audit is also considered as the key component of clinical governance. Clinical audit is a tool which can be used to realize how the clinical care is being provided against the explicit criteria and to learn if there are opportunities for improvement. Clinical audit may be used to improve aspects of care in a wide clinical area. It can also be used in association with changes in care provision or to confirm that current practice meets the expected level of performance.

Clinical audit has many advantages:

- It offers a way to assess and improve patient care
- Helps to uphold professional standards
- Help healthcare staff to identify and measure areas of risk within their service
- Helps to create a culture of quality improvement in the clinical setting
- Used as educational platform for the participants
- It involves being up to date with evidence based good practice
- It offers an opportunity for increased job satisfaction
- It is increasingly seen as an essential component of professional practice
- It can improve the quality and effectiveness of healthcare service

1.2 Scope of the manual

The aim of this document is to prepare organized training manual for healthcare providers and healthcare service administrators in Ethiopia to enable them aware of clinical audit concept, process of clinical audit, support best process in clinical audit, improve awareness of clinical audit as an integral part of clinical practice. Thus, this

training course manual will provide practical guide to the methodological aspect of clinical audit including the wider considerations for ethical/data protection/confidentiality primarily for healthcare professionals and healthcare service administrators involved in clinical service currently and used as a reference during and after training delivered.

It is trusted that the module is clinically relevant and will improve the practice of clinical service. It is educationally all-encompassing, recognizing the specific needs of the adult learner. In keeping with this, it will comprise a range of teaching modalities to include skill sessions, scenarios, demonstrations, discussions and interactive lectures.

1.3 Core competency of the training

- After completion of this module, the participants will be able to carry out clinical audits on particular service areas in order to improve healthcare delivery

1.4 Rationale of clinical audit training

There is a considerable quality gap in the delivery of health care along one or more quality dimensions, including people-centeredness, safety, timeliness, effectiveness, efficiency, equity, and integration, according to a growing body of research. A well-organized, efficient clinical auditing program must be implemented as one of the measures taken to increase the quality and safety of the care given.

In order for clinical audit programs in healthcare facilities to be successful, two requirements must be met: the use of appropriate methodologies and creating a supportive environment.

As a result, there are problems with the application of appropriate methods in terms of careful planning, creating simple and practical audit standards and criteria, designing and monitoring appropriate quality improvement plans based on identified gaps, connecting audit findings with quality improvement projects (systematic management of change), as well as the absence of an appropriate structure that can organize and provide the necessary support for auditors in developing their audit project.

This document describes the idea of clinical audit, the procedures required in performing a clinical audit, the roles and duties of key participants, and techniques for monitoring the advancement of clinical audit projects. Additionally, to make it easier to conduct clinical audits regularly in health facility, simpler clinical audit tools on certain topic areas based on the clinical guidelines and protocols.

2 Module syllabus

2.1 **Module name**: Clinical audit

2.2 **Module code**: 4 days

2.3 **Module duration**: 5 days

2.4 **Pre-requisite:** None

2.5 **Module continuing education unit (CEU)-** 15 CEU

2.6 **Module description**: This is a five days training module/course is designed for clinicians, policy makers, healthcare service leaders and administrators to equip with all the essential knowledge, skills and attitudes on Identifying areas of health care which needs improvement, conducting clinical audit on specific healthcare area and the changes after conduct of clinical audit. It focuses on the health service quality improvement activities.

2.7 **Module/course goal (CC)**:

- After the completion of this module, participants will be able to conduct clinical audits on particular service areas in order to improve healthcare delivery.

2.8 **Enabling/learning objectives**

At the end of this training the trainees will be able to:

- Identify principles of clinical audit
- Plan and prepare for audit
- Identify best practice (standards)
- Mobilize and map resources to conduct clinical audit
- Measure level of performance
- Manage changes for healthcare delivery and maintain sustainability

2.9 **Chapters/sessions outlines**

Chapter	Title /topic	Durations	Sessions
1.	Introduction to clinical audit	1 hour	Session I: introduction Session II: audit cycle Session III: importance of clinical audit Session IV: difference between clinical audit, research and service evaluation
2.	Planning For Audit	2 hours	Session 1: Session 2:
3.	Identify the Best Practice	7 hours	
4.	Resource mobilization and mapping	3 hours	
5.	Measuring Level of Performance:	7 hours	
6.	Making Improvements	4 hours	
7.	Sustaining Improvements	3 hours	

2.10 **Teaching learning methods and activities**

- Interactive lectures
- Small Group work, reporting and feed back

- Role play
- Case studies
- Guided activity of using online resources
- Feedback on self-reflection

2.11 **Learning materials and resource (including references)**

Training materials

- Lap top computer
- LCD projector
- White board
- White board marker
- Flip chart
- Permanent marker
- Note book
- Pen
- Sticker notes

2.12 **Learning Assessment methods**

- Written exam
- Observation of case study analysis in small groups
- Observation of group activity/ audit proposal presentation
- Review of the completed clinical audit document

2.13 **Criteria for course completion and Criteria for certification /earning CEU**

To earn 15 CEU after completion of training, the trainees must score 80 and above points from the following assessments.

- Written test (20%)
- Observation of small group case study analysis (10%)
- Observation of group presentation of clinical audit proposal (20%)
- Review of completed individual clinical audit document (50%): This should be conducted as a home take project after the first phase of the training and submitted on the agreed time period set during the training or at the end of the training based on the ready-made data given to the trainees.

13. Trainer and facilitators selection criteria

- have basic/TOT on clinical audit training or course
- participate on clinical audit training module preparation

14. Trainees and participants selection criteria

- all healthcare professionals currently engaged in clinical service
- health care administrators/leaders and healthcare policy makers

15. Training evaluation methods (satisfaction, learning, behavior and impact)

- trainees' reaction using daily evaluation and end of training evaluation

- direct observation of trainees' behaviors
- interview of trainees and facilitators

16. Venue:

Accredited CPD provider center

17. Composition

Maximum of 20 participants from health institution (teaching/non-teaching)

Physicians, anesthetists, pharmacists, healthcare leaders (CCD, Quality directors)

Number of trainers: 4, number of facilitators: 3

18. Schedule

Day 1			
Time	**Session**	**Facilitator**	**Moderator**
8:00-8:30	Registration	-	-
8:30-8:45	Welcoming and introduction	-	-
8:45-9:00	Pretest	-	-
9:00:9:45	Chapter 1: planning for audit - Introduction to clinical audit - Audit cycle - Audit topic selection	-	-
9:45-10:15	- Group activity (select audit area/topic)	-	-
10:15:10:45	Tea-break		
10:45-12:30	- Discussion on group activity (Select prioritize audit topic)	-	-

Lunch			
2:00- 2:30	Chapter 2: Identifying best clinical practice (setting audit standards)		
2:30-2:45	- Group activity (setting standards from different sources for the area identified)		
2:45-3:15	TB		
3;15-5:30	Discussion and reflection of group works		

Day 2			
Time	**Session**	**Facilitator**	**Moderator**
8:30-8:45	Reflection and recap of day one session	-	-
8:45-9:45	Chapter 2: continued - Sources of clinical audit standards	-	-
9:45:10:15	- Group activity on the developing audit standard using different sources	-	-
10:15:10:45	Tea-break		
10:45-12:30	- Discussion on group activity	-	-
Lunch			
2:00- 2:45	Chapter 3: resource mobilization and mapping		

2:45-3:15	TB		
3:15-3:45	- Group activity (identifying resources and stalk holders for clinical audit)		
3;45-5:30	- Discussion and reflection of group works		

Day 3			
Time	**Session**	**Facilitator**	**Moderator**
8:30-8:45	Reflection and recap of day two session	-	-
8:45-9:45	Chapter 4: measuring level of performance	-	-
9:45:10:15	- Group activity on the designing audit data collection tool (audit proforma)	-	-
10:15:10:45	Tea-break		
10:45-12:30	- Discussion on group activity	-	-
	Lunch		
2:00- 2:30	Chapter 4: measuring level of performance continued		
2:30-2:45	- Group presentation of the audit plan with planner		
2:45-3:15	Tea break		

3;15-5:30	Group presentation of the audit plan with planner continued		

Day 4			
Time	**Session**	**Facilitator**	**Moderator**
8:30-8:45	Reflection and recap of day three session	-	-
8:45-9:45	Chapter 5: making improvements - Agreed actions - Initiating changes	-	-
9:45:10:15	- Change management	-	-
10:15:10:45	Tea-break		
10:45-12:30	- Group activity on how to initiate changes from the audit result	-	-
	Lunch		
2:00- 2:45	Chapter 5: making improvements continued		
2:45-3:15	Tea break		
3;15-5:30	Group presentation of the audit plan with planner		

Day 5			
Time	**Session**	**Facilitator**	**Moderator**
8:30-8:45	Reflection and recap of day four session	-	-
8:45-9:45	Chapter 6: sustainability of improvements	-	-
9:45:10:15	- Reaudit	-	-
10:15:10:45	Tea-break		
10:45-12:30	- Group discussion on the challenges of reaudit to ensure sustainability of improvements/changes	-	-
Lunch			
2:00- 2:45	Post test		
2:45-3:15	Tea-break		
3;15-5:30	Final training evaluation, certification and closing ceremony		

Chapter 1: Principles of clinical audit

Duration: 5½ Hours

Primary objective:

At the end of this chapter, participants will be able to analyze principles of audit and audit cycle to conduct the quality improvement clinical audit.

Enabling objectives:

At the end of this session, participants will be able to:

- Describe clinical audit
- Describe the importance of conducting clinical audit
- Discuss the difference between clinical audit, research and service evaluation
- Explain types of clinical audit
- Evaluate the stages of audit cycles

Summary of activities

No	Activity	Method of delivery	Time	Slide Number	Resource
1.	Introduction of Clinical Audit	• Question and Answer • Group Activity • Interactive Lecture and discussion	15 minutes 15 minutes		LCD Flip Chart Computer markers
2.	Audit cycle	• Group Discussions • Interactive Lecture	15 min		
3.	Types of Clinical Audit	• Group Discussions • Think-Pair-Share • Think-Pair-Share	45 min		
4.	Difference between Research, Clinical Audit and Service Evaluation	• Gallery Walk • Interactive Lecture	40 min		

Learning activities: Introduction to clinical audit

	Topic	Time	Activity
1	Introduction of Clinical Audit	5 min	Question and Answer • Read the question to the whole participants • Allow the participants to think on the question • Reflect for the whole group • Validate and Summarize the answers for question Group Discussion • Read the case scenario • Divide into groups of 3 and brain storm • Discuss with the whole group

2	Importance of Clinical	5min	Group Discussions • Read the case scenario • Divide into groups of 3-5 and brain storm • Discuss with the whole group
4	Types of Clinical Audit	2 min	Group Discussions • Read the case scenario • Divide into groups of 3-5 and brain storm • Discuss with the whole group Think-Pair-Share • T: (Think) Ask a specific question about the text. Participants "think" about what they know or have learned about the topic. • P: (Pair) Each participants should be paired with another participants or a small group. • S: (Share) Participants share their thinking with their partner. Facilitators expand the "share" into a whole-class discussion.
5	Difference between Research, Clinical Audit and Service Evaluation	3 min	Instruction to conduct Gallery Walk ❖ Provide the assignment ❖ Prepare the space ❖ Prepare the materials ❖ Introduce the activity ❖ Divide into groups ❖ Allow time for discussion Wrap up

Chapter two: Clinical audit proforma

Duration: 8 ½ Hours

Primary objective:

At the end of this chapter participants will be able to demonstrate comprehension of the audit planning process, audit methodology and clinical audit tool.

Enabling objectives:

At the end of this session, participants will be able to:

- Select appropriate audit topic based on topic selection criteria
- Set appropriate clinical audit aim and objectives
- Analyze the audit standard by identifying best practices in the area
- Construct audit methodology using audit planner
- Develop organized clinical audit tool

Summary of activities

No	Activity	Method of delivery	Time	Slide Number	Resource
1.	Planning for audit	• Question and Answer • Group Activity • Interactive Lecture and discussion	10 min 30 min		LCD Flip Chart Computer markers
2.	Prioritizing audit topics	• Group Discussions • Interactive Lecture	30 min		
3.	Aim and objectives	• Gallery-walk • Interactive lecture	20 min		
4.	Audit standards	• Group activity • Guided practice • Interactive Lecture	20 min 30 min		
5.	Resource mobilization and mapping	• Think-pair-share • Interactive lecture and discussion	15 min		
6.	Audit methodology	• Question and answer • Group work and presentation	20 min 120 min		

Learning activities

	Topic	Time	Activity
1	Planning for audit	30 min	Question and Answer • Read the question to the whole participants • Allow the participants to think on the question • Reflect for the whole group • Validate and summarize the answers for question Group Discussion • Read the case scenario • Divide into groups of 3 and brain storm • Discuss with the whole group

2	Prioritizing audit topics	30 min	Group Discussions • Read the case scenario • Divide into groups of 3 and brain storm • Discuss with the whole group
3	Aim and objectives	20 min	Instruction to conduct Gallery Walk ❖ Provide the assignment ❖ Prepare the space ❖ Prepare the materials ❖ Introduce the activity ❖ Divide into groups ❖ Allow time for discussion ❖ Wrap up
4	Audit standards	20 min	Group Discussions • Read the case scenario • Divide into groups of 3-5 and brain storm • Discuss with the whole group
5	Resource mobilization and mapping	15 min	Think-Pair-Share • T: (Think) Ask a specific question about the text. Participants "think" about what they know or have learned about the topic. • P: (Pair) Each participants should be paired with another participants or a small group. S: (Share) Participants share their thinking with their partner. Facilitators expand the "share" into a whole-class discussion.
6	Audit methodology	20	Question and Answer • Read the question to the whole participants • Allow the participants to think on the question • Reflect for the whole group • Validate and summarize the answers for question Group Discussions and presentation • Give the assignment to work in group • Divide into groups of 3-5 and brain storm • Allow them to present in their larger group • Discuss with the whole group

Chapter 3: Measure level of performance and making improvements

Duration: 10 ½ hours

Primary objective: At the end of this session, participants will be able to make and maintain improvement in healthcare delivery by conducting clinical audit.

Enabling objectives:

At the end of this chapter, participants will be able to:

- Analyze types of data and data collection
- Interpret audit data and compare with agreed standards
- Draw conclusion from results of clinical audit data
- Plan for change implementation
- Apply results of audit for improving quality of service
- Conduct re-audit

Summary of activities

No	Activity	Method of delivery	Time	Slide Number	Resource
1.	Data collection	• Question and Answer • Group Activity • Interactive Lecture and discussion	10 min 30 min		LCD Flip Chart Computer markers
2.	Data analysis and drawing conclusion	• Think-pair-share • Interactive Lecture • Case study	100 min		
3.	Making changes and improvements	• Write pair share • Interactive lecture and discussion • Group activity	20 min		
4.	Sustainability of changes	• Think-pair-share • Interactive Lecture	10 min		
5.	Performance indicators	• Role play • Interactive lecture and discussion	20 minutes		

Learning activities

	Topic	Time	Activity

1	Data collection	40 min	Question and Answer • Read the question to the whole participants • Allow the participants to think on the question • Reflect for the whole group • Validate and summarize the answers for question Group Discussion • Read the case scenario • Divide into groups of 3 and brain storm • Discuss with the whole group
2	Data analysis and drawing conclusion	30 min	Case study • Present the case of clinical audit for discussion in their group • Work in their group and present their ideas • Reflect the group idea for the class and discuss • Windup of the case study Think-Pair-Share • T: (Think) Ask a specific question about the text. Participants "think" about what they know or have learned about the topic. • P: (Pair) Each participant should be paired with other participants or a small group. • S: (Share) Participants share their thinking with their partner. Facilitators expand the "share" into a whole-class discussion.
3	Making changes and improvements	15 min	Think- pai- share • T: (Think) Ask a specific question about the text. Participants "think" about what they know or have learned about the topic. • P: (Pair) Each participants should be paired with another participants or a small group. • S: (Share) Participants share their thinking with their partner. Facilitators expand the "share" into a whole-class discussion. Group Discussions • Read the case scenario • Divide into groups of 3-5 and brain storm • Discuss with the whole group
4	Sustainability of changes	10 min	Think-Pair-Share • T: (Think) Ask a specific question about the text. Participants "think" about what they know or have learned about the topic. • P: (Pair) Each participants should be paired with another participants or a small group. • S: (Share) Participants share their thinking with their partner. Facilitators expand the "share" into a whole-class discussion.
5	Performance indicators	15 min	Role play •

Annexes

Annex I: Clinical Audit proforma

Clinical Audit Planning & Summary Form	Federal Ministry of Health Health Service Quality Directorate

Project Title: (Identified Audited practice to Improve surgical access – at least three topics)	Audit Ref No

Project Team	
1. Senior Clinician	**2. Project leader (junior person doing the data collection)**
3. Name of person from the audited department	4. Others, as necessary

Background

What?

Why?

What benefits for the patient?

Aims& Objectives

Standards	Target	Evidence	Data source

Clinical Audit Planning & Summary Form	Federal Ministry of Health Health Service Quality Directorate

Project Title: (Identified Audited practice to Improve surgical access – at least three topics)	Audit Ref No

Audit Methodology

Audit Population

Audit Sample

Data collection method

Deadlines

Start date

Completion date

Presentation date

Main Results (compared to specific standards)

Clinical Audit Planning & Summary Form	Federal Ministry of Health Health Service Quality Directorate
Project Title: (Identified Audited practice to Improve surgical access – at least three topics)	**Audit Ref No**
Conclusions (highlight areas identified for improvement) Areas of good practice Areas which need to be improved	

Action to be implemented	**By whom**	**By When**

Clinical Audit Planning & Summary Form		Federal Ministry of Health Health Service Quality Directorate
Project Title: (Identified Audited practice to Improve surgical access – at least three topics)		**Audit Ref No**
Re-audit required:- Yes / No		
Presentation **Date**		

Annex II: Gallery Walk

A gallery walk is a great way to engage participants in a training session. It can be used to introduce a new topic, review material, or even to assess understanding. Here are some instructions for conducting a successful gallery walk:

1. Prepare the space: Before the gallery walk, make sure the space is set up for success. Arrange the room so that participants can easily move around and view the materials. If possible, provide tables or stands to display the materials.

2. Prepare the materials: Gather the materials that you want to use for the gallery walk. These can include posters, handouts, pictures, videos, or any other visual aids. Make sure the materials are clearly labeled and organized in a way that makes sense.

3. Introduce the activity: Explain to the participants what a gallery walk is and how it will be conducted. Let them know that they will be moving around the room to view the materials and that they should take notes or ask questions as they go.

4. Divide into groups: Divide the participants into small groups and assign each group to a specific area of the room. Make sure each group has enough time to view the materials and discuss them.

5. Allow time for discussion: After each group has had a chance to view the materials, allow time for discussion. Encourage participants to ask questions and share their thoughts and ideas.

6. Wrap up: At the end of the gallery walk, review the materials and discuss any key points or takeaways. Make sure to thank the participants for their participation and encourage them to ask any remaining questions.

By following these instructions, you can ensure that your gallery walk is a successful and engaging activity for your training session. It can be a great way to introduce a new topic, review material, or assess understanding

Annex III: Tips for successful development of clinical audit

Top tips for the successful development of clinical audit data collection tools (*Developed by the Irish Clinical Audit Network 2006)*

Validity As in research, the data collection tool should have a high level of 'validity' i.e. it collects the data that it purports to collect. This is why it is important to start with clear clinical audit objectives and use evidence-based standards from the outset.

Reliability As in research, the data collection tool should have a high level of reliability, i.e. different personnel collecting the data will record the exact same or highly similar data using the tool. Inter-rate reliability: If data is being collected by many personnel it is important that a check is made to ensure that inter-rater reliability is high.

Sequencing Data collection questions should flow logically based on the sequence of events in the data source i.e. patient records etc. If feedback is sought from service users or staff members questions should be worded clearly and follow the same basic principles used in developing research interview schedules.

Quantitative versus qualitative clinical audit data collection Clinical audits can use either quantitative or qualitative data to meet the clinical audit objectives.

- The majority of clinical audits focus on quantitative data. Data collection tools of this type should ensure that clear closed questions are used i.e. Yes/No, Documented/Not documented etc.

- Where qualitative data is required it is necessary to develop data collection questions for use in interviews/focus groups or via questionnaire.

Drafting closed questions When drafting closed questions:

- use mutually exclusive options, for example Yes/No; Male/Female

- include an option for all responses, for example Yes/No/Don't know/Not applicable

- where appropriate, give more than two options, for example, 4 or 6 (an even number is better as this may assist selection of the middle option as a default). Reliability tends to rise as the number of options rise.

Presentation Ensure that the data collection tool is present in such a way that it is visually appealing to those collecting the data:

- Use standard fonts.

- Ensure that the font size is readable.

- Do not overcrowd each page – ensure that there is space for recording all relevant information and additional comments.

Comments Ensure that there is sufficient space for including comments during data collection Include the date of data collection Ensure that the date of data collection is recorded clearly on each data collection sheet. This is particularly helpful when data is collected over an extended timeframe.

Annex IV: Think-pair-share

Name _____ Date_____

Think-Pair-Share

Read the following question or problem:

Think

On your own, write three ideas you have about this question or problem:
1._____
2._____
3._____

Pair

Discuss your ideas with a partner. Put a check by any ideas, above, that your partner also wrote down. Then, write down ideas your partner had that you did not have:
1._____
2._____
3._____

Share

Review all of your ideas and circle the one you think is most important. One of you will share this idea with the whole group. As you listen to the ideas of the whole group, write down three more ideas you liked:
1._____
2._____
3._____

Annex V: sample clinical audit for case study

An audit of the knowledge of 5 key National Institute of Health and Clinical Excellence (NICE) clinical guidelines relevant to Health Visiting and School Nursing practice- 2012/13 (436)

Audit period: September 2012-November 2012

Report date: February 2013

Executive Summary

Why the audit was undertaken

The overall aim of this audit was to ensure that health visitors and school nurses in FYPC have a working knowledge of 5 key NICE guidelines relevant to practice that have been published between 2007 and 2010. In order to achieve this it was necessary to establish the level of knowledge in 3 areas for each of the conditions:

Key signs and symptoms

When further action is needed

The correct management and advice to give.

A similar audit had been undertaken in the previous county organisation which highlighted concerning gaps in knowledge, although this was considered to be a new baseline audit as it used a different tool and methodology, and was carried out over a new organisation.

How the audit was carried out

A questionnaire was designed with questions around each of the above areas of knowledge, asking for one response per question. Staff were asked to complete the questionnaire at professional meetings, where attendance was expected to be high. They had no prior warning of the audit.

Key findings

A number of areas of concern were identified, with compliance being rated as minimal in accordance with LPT clinical audit policy (dated April 2012) Some of these need addressing as they indicate practice which is contrary to that recommended in the NICE guidance as well as in the Standard Operating Procedures for health visitors. Full analysis of the results was made difficult by the high number of multiple responses. It also became clear that for the school nurses, compliance was below that originally expected for some conditions, but as these related to situations they would not meet on a regular or frequent basis it could be argued that these should not have been included. However, a surprisingly low number of staff said they were aware of where to access NICE guidelines, despite all having access to the internet.

Key actions

The key recommendations arising from the results of the audit of the health visitors are around education and raising awareness of correct diagnosis and management of jaundice and of eczema. Some actions, such as training and circulation of new operating procedures, have started to take place and will go some way towards addressing the knowledge gaps in these areas. Further learning points will be disseminated by line managers to individual teams and discussed at reference group meetings.

Re-audit date

The audit will be repeated between September and November 2013.

Aim

The overall aim of the audit is to ensure that health visitors and school nurses in the city and county localities have a working knowledge of 5 key NICE guidelines relevant to practice.

Objectives

The objectives are:

to establish whether health visitors and school nurses are aware of the key signs and symptoms of the condition specified in each guideline.

to establish whether they can identify when further action is needed.

to establish whether they are aware of the correct management and advice to give.

Criteria & standards

The following NICE guidance were audited:

National Institute for Health and Clinical Excellence

CG 47 Feverish Illness in children under 5 (May 2007)

CG57 Atopic Eczema in Children (December 2007)

CG84 Diarrhoea and vomiting in children under 5 (April 2009)

CG98 Neonatal jaundice (May 2010)

CG99 Constipation in children and young people (May 2010)

None of the above is in the process of being updated, although CG47 is scheduled for review in May 2013.

Table 1 - Criteria and standards

Criteria	Standards
Health visitors and school nurses can correctly identify what NICE stands for, and where to access the clinical guidelines.	The expected standard is 100%.

Questions 1 and 17. All participants in the audit are registered nurses and should be aware of current guidelines and where to access them as part of their professional code of conduct.	
Health visitors and school nurses can make a correct diagnosis, identify when further action is needed and identify what the correct management advice is for the following 4 conditions: diarrhoea and vomiting in children under 5, feverish illness in children under 5, atopic eczema and constipation. Questions 2,3,4and 8-16,	The expected standard is 95%.

Although this is less than the 100% compliance expected for each of the guidelines, this will allow for practitioners who do not use this knowledge on a daily basis and may need to seek further advice themselves before offering advice to clients. It will also allow for practitioners who are new in post and who will not have received emails raising awareness of NICE guidelines over recent months.

Health visitors can correctly identify key clinical signs of prolonged neonatal jaundice, when further action is needed and what the correct management is. Questions 5,6 and 7,	The expected standard is 100%.

As the NICE guideline audit tool sets compliance at 100% then the standard of knowledge must also be 100%. This is an area of practice which health visitors could encounter on a daily basis while visiting children at home, so a sound working knowledge is essential.

The following rating was applied to the results to highlight any particular areas of concern, which is consistent with the trust agreed approach and the recommendation of the Clinical Audit Support Centre.

Full compliance	Partial compliance	Minimal compliance
$x \geq 95$ (or 100)%	70%$\leq x <$94(or 99)%	$x <$ 69%

Method

Questionnaires were completed by staff during professional locality meetings from September to November 2012. The audit was listed as an item on the agenda for team meetings as NICE but there was no pre-warning about the audit. Staff were given time to complete the questionnaire at the beginning of the meeting, ensuring a high response rate, and handed them straight back to the manager who sent them to the audit team for analysis. Locality and team meetings are seen as a priority for all health visiting and school nursing staff so a high representative sample should have been achievable. Completed audit forms were returned to the audit office and were inputted and analysed using Excel.

Sample & data source

A minimum target of 160 would have given a confidence level of 95% with a 5% error margin. The total population is approximately 270.

Audit type

This is a new baseline audit.

See **Error! Reference source not found.** on p.**Error! Bookmark not defined.** for a copy of the audit tool.

Service areas / teams included

The audit was carried out on teams from health visiting and school nursing across all localities in the city and the county. It had been aimed at qualified members of the team (i.e. nurses registered with the NMC) but some were completed by assistants within the teams and by staff with nursery nurse qualification. It was decided to exclude these results from the final analysis as the expected compliance for this group of staff would be different, due to the differing roles and responsibilities. Registered nurses working within health visiting teams were included.

Although the questionnaire asked for locality to be identified this proved to be problematic when analysing the results due to differing terminology used. For example there are two North West and two South teams, one each in the county and the city. It was decided therefore to analyse results without taking locality or neighbourhood into account as the expected compliance was the same across the whole area

Findings

Responses received

A total of 147 questionnaires were received from qualified staff, 111 from health visiting teams and 36 from school nurse teams. This was lower than expected but could be accounted for by anecdotal reports of team managers not being informed about the audit, lack of mutually convenient times for the audit team to attend the team meetings, and some teams declining to participate. Two questionnaires were returned with all answers blank, and were therefore excluded, as were those which did not indicate which professional group they belonged to. Some questionnaires were completed by community nursery nurses but these were not included in the analysis as the audit was looking at the knowledge of NMC registered nurses. This response rate still gives a confidence level of 95% with a margin of error of 7.15%.

Audit findings

Although it had originally been planned for the questionnaires to be electronically analysed on FORMIC this proved to be problematic due to the high numbers of staff ticking more than one box for the answer. Each question then had to be looked at to decide clinically whether to count the answer as right or wrong, depending which boxes were ticked. For example, in question 7, if someone ticked A or B as well as the correct answer, C, then they would be counted as correct, because the advice in A and B could also form part of the overall management, but would not be correct on their own. Similarly, if someone ticked an answer which was definitely incorrect as well as the correct one then they would be marked wrongly

Table 2 Audit results for health visiting teams (in comparison with thresholds of compliance)

Criteria	Expected standard	Number answered correctly	Compliance
Question 1: NICE stands for National Institute for Health and Clinical Excellence	100%	62	56%
Question 2: Gastroenteritis alone should be suspected if there is a sudden change to loose or watery stools.	95%	106	95%
Question 3: Which if the following children with gastroenteritis can be safely managed at home? An 18 month old child who has had diarrhoea for 6 days.	95%	93	84%
Question 4: Which of the following is the correct advice to a mother of a 1 year old with gastroenteritis, no risk factors and no signs of dehydration? Encourage breastfeed/other milk feeds and fluid intake, advice about Red Flag symptoms and how to get immediate help if they develop.	95%	99	89%
Question 5: Which of the following is NOT a sign of prolonged jaundice? Yellow colouration of blanched skin in a term baby aged 10 days.	100%	84	76%
Question 6: Which of the following will raise your concern about a baby with jaundice? Dark urine which stains the nappy.	100%	107	96%
Question 7: If a baby has prolonged jaundice at the primary visit which of the following described the correct management? Contact children's day ward at the LRI to make a referral, arrange the appointment and inform the parent(s).	100%	89	80%
Question 8: Which of the following can be used to diagnose a feverish illness? Parental report of a fever.	95%	15	14%
Question 9: Which of the following is a Red Flag symptom in a child with feverish illness (which would cause you to refer to a paediatric specialist urgently)? Weak or high pitched cry.	95%	104	94%
Question 10: Which of the following is a correct piece of advice for parents of a child with a fever (with no red or amber symptoms)? Look for signs of dehydration, non-blanching rash, encourage more fluids than usual and to seek further advice if they feel the child is getting worse.	95%	99	89%

Criteria	Expected standard	Number answered correctly	Compliance
Question 11: A diagnosis of atopic eczema can be made when a child has an itchy skin condition and which of the following? History of dry skin in the last 12 months, visible flexural dermatitis involving skin creases and allergic rhinitis.	95%	94	84%
Question 12: A referral to specialist dermatological advice is necessary if: Atopic eczema is associated with severe or recurrent infections.	95%	106	95%
Question 13: Which of the following describes the correct management for a 2 year old with dry skin, and occasional appearance of red patches of itchy skin behind the knees? Emollients, mild topical corticosteroids only where and when the skin is red and itchy, and continue with emollients even when fully resolved.	95%	85	77%
Question 14: Which of the following children is constipated? A 3 year old who passes 2 hard large stools per week, straining to do so and who has had constipation in the past.	95%	110	99%
Question 15: When assessing a child, which of the following findings is a Red Flag indicating an underlying disorder rather than idiopathic constipation? Ribbon-shaped stool and abdominal distension with vomiting.	95%	103	93%
Question 16: Which of the following describes the most appropriate dietary and lifestyle advice for parents of a 5 year old child with idiopathic constipation (in addition to management with laxatives)? Fluid intake of at least 1200ml as drinks daily, recommend foods such as fruit and vegetables, high-fibre bread and wholegrain cereals, daily physical activity and negotiated follow-up and support.	95%	105	95%
Question 17: Are you aware of where to access the NICE guidelines quick reference guides? Yes	100%	94	85%

Table 3 Audit results for school nursing teams (in comparison with thresholds of compliance)

Criteria	Expected standard	Number answered correctly	Compliance
Question 1: NICE stands for National Institute for Health and Clinical Excellence	100%	24	67%
Question 2: Gastroenteritis alone should be suspected if there is a sudden change to loose or watery stools.	95%	29	81%
Question 3: Which if the following children with gastroenteritis can be safely managed at home? An 18 month old child who has had diarrhoea for 6 days.	95%	25	70%
Question 4: Which of the following is the correct advice to a mother of a 1 year old with gastroenteritis, no risk factors and no signs of dehydration? Encourage breastfeed/other milk feeds and fluid intake, advice about Red Flag symptoms and how to get immediate help if they develop.	95%	24	67%
Question 8: Which of the following can be used to diagnose a feverish illness? Parental report of a fever.	95%	2	6%
Question 9: Which of the following is a Red Flag symptom in a child with feverish illness (which would cause you to refer to a paediatric specialist urgently)? Weak or high pitched cry.	95%	22	61%
Question 10: Which of the following is a correct piece of advice for parents of a child with a fever (with no red or amber symptoms)? Look for signs of dehydration, non-blanching rash, encourage more fluids than usual and to seek further advice if they feel the child is getting worse.	95%	24	67%
Question 11: A diagnosis of atopic eczema can be made when a child has an itchy skin condition and which of the following? History of dry skin in the last 12 months, visible flexural dermatitis involving skin creases and allergic rhinitis.	95%	34	94%
Question 12: A referral to specialist dermatological advice is necessary if: Atopic eczema is associated with severe or recurrent infections.	95%	34	94%
Question 13: Which of the following describes the correct management for a 2 year old with dry skin, and occasional appearance of red patches of itchy skin behind the knees? Emollients, mild topical corticosteroids only where and when the skin is red and itchy, and continue with emollients even when fully resolved.	95%	23	64%
Question 14: Which of the following children is constipated? A 3 year old who passes 2 hard large stools per week, straining to do so and who has had constipation in the past.	95%	34	94%

Criteria	Expected standard	Number answered correctly	Compliance
Question 15: When assessing a child, which of the following findings is a Red Flag indicating an underlying disorder rather than idiopathic constipation? Ribbon-shaped stool and abdominal distension with vomiting.	95%	35	97%
Question 16: Which of the following describes the most appropriate dietary and lifestyle advice for parents of a 5 year old child with idiopathic constipation (in addition to management with laxatives)? Fluid intake of at least 1200ml as drinks daily, recommend foods such as fruit and vegetables, high-fibre bread and wholegrain cereals, daily physical activity and negotiated follow-up and support.	95%	36	100%
Question 17: Are you aware of where to access the NICE guidelines quick reference guides? Yes	100%	29	81%

How compliance was calculated

$$\text{Compliance} = \frac{\text{number of YES}}{\text{number of YES and NO and BLANK}} \times 100$$